

PRACTICAL - 9

Aim: Cleaning and Structuring Data for Analysis

- Identify and handle missing or inconsistent data
- Convert wide data into tall format
- Perform union of multiple files
- Execute cross-database joins
- Restructure data for better visualization
- Handle different levels of detail in datasets
- Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data

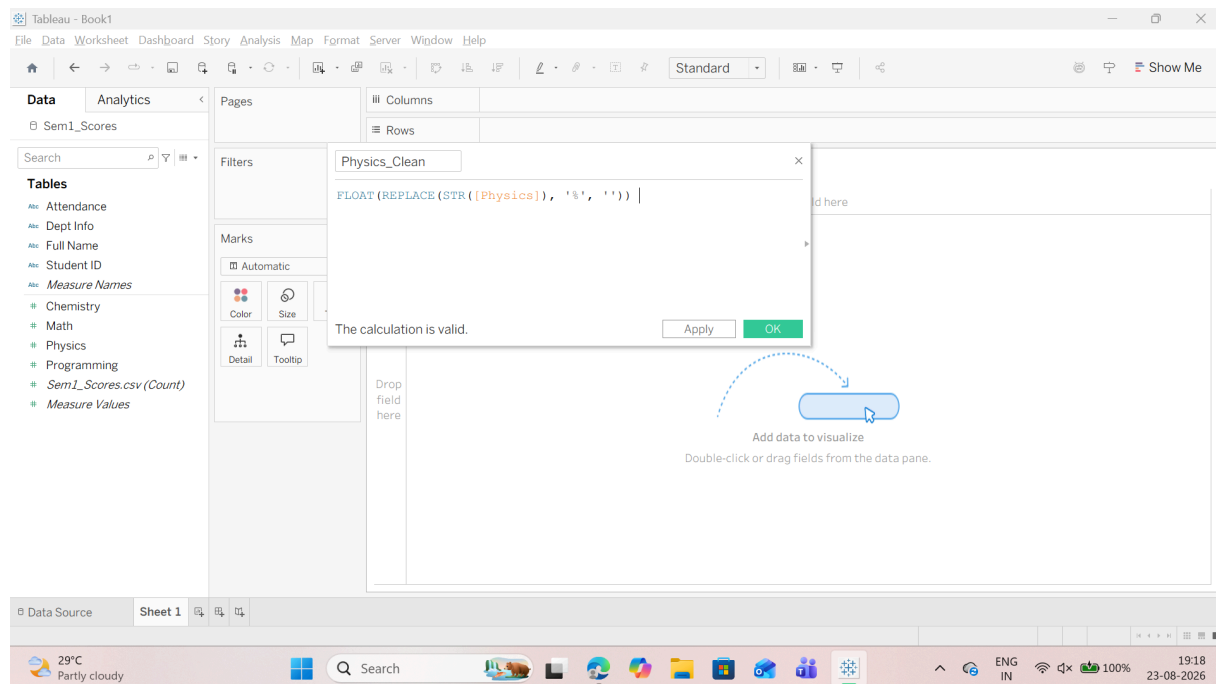
Open **Sem1_Scores.csv** in Tableau using **Text File**, then check the Data Source preview to identify **Null values in Math, Physics, and Chemistry**.

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. The 'Data Source' tab is active, displaying a preview of the data. The preview table has the following structure:

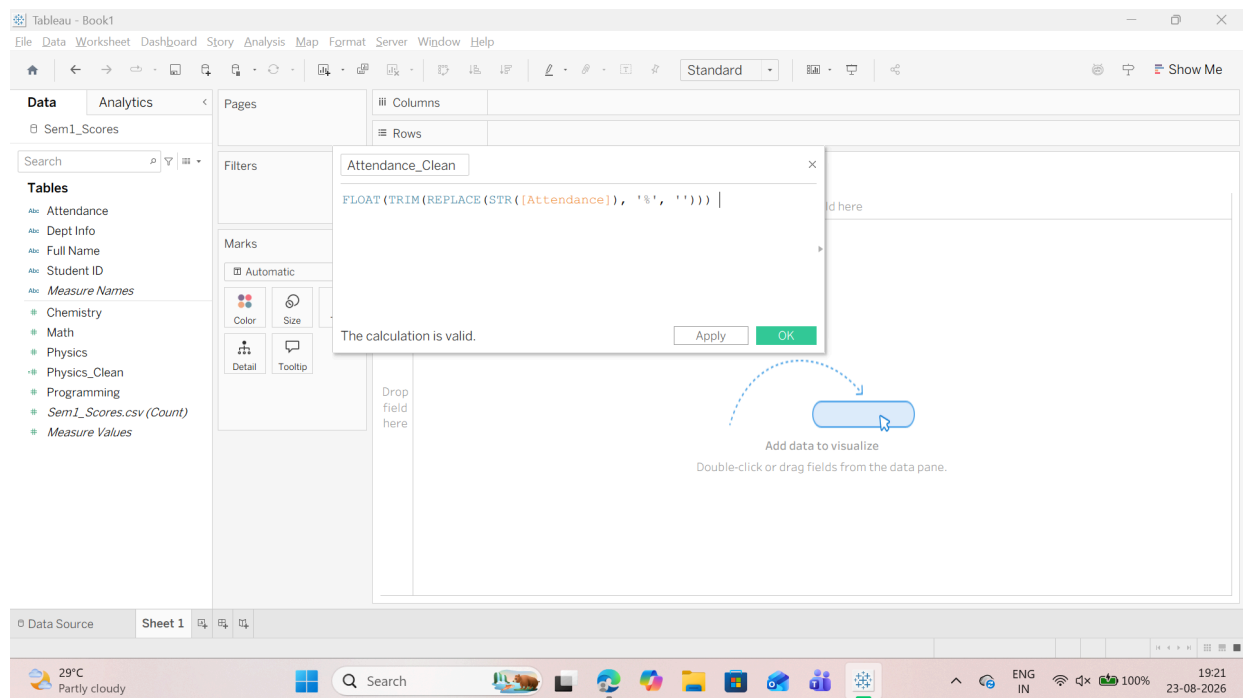
Student ID	Full Name	Dept Info	Math	Physics
S206	Riya Kapoor	CSE-01	87	0.900000
S207	Manav Shah	ECE-02	null	0.740000
S208	Tanya Rao	IT-03	93	0.960000
S209	Dev Malhotra	CSE-01	71	0.650000

The 'Math' column shows a null value for student S207. The interface also shows the 'Connections' pane on the left with 'Sem1_Scores' selected, and the 'Fields' pane in the center showing the available fields for the data source.

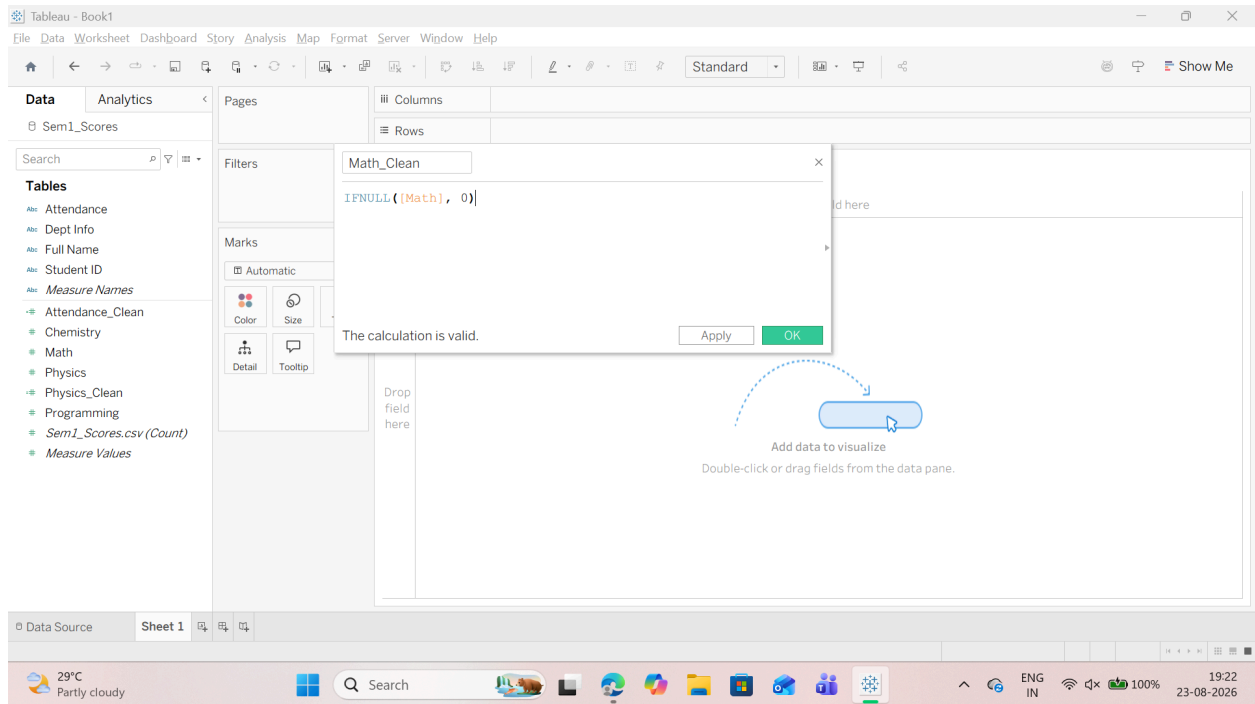
In **Sheet 1**, create a calculated field named **Physics_Clean** using **FLOAT(REPLACE(STR([Physics]), '%', ''))** to convert Physics text percentages into numeric values.



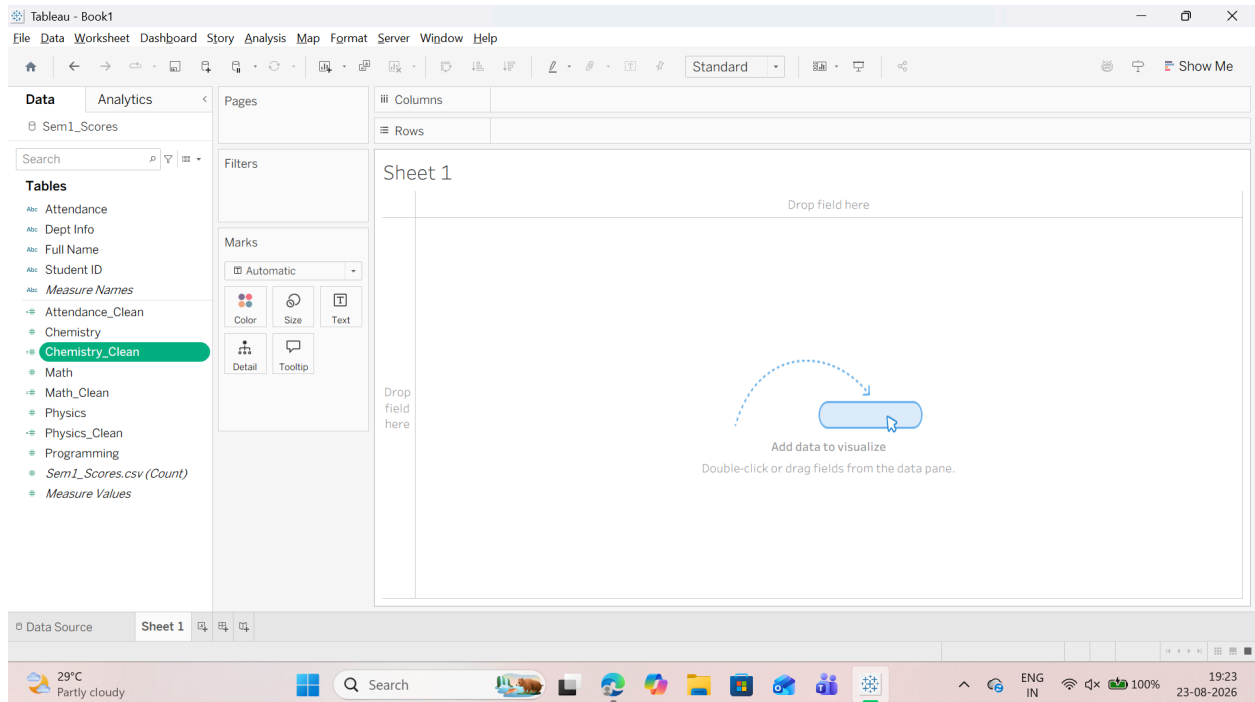
Create **Attendance_Clean** with **FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))** to convert Attendance text percentages into numeric values, then click **OK**.



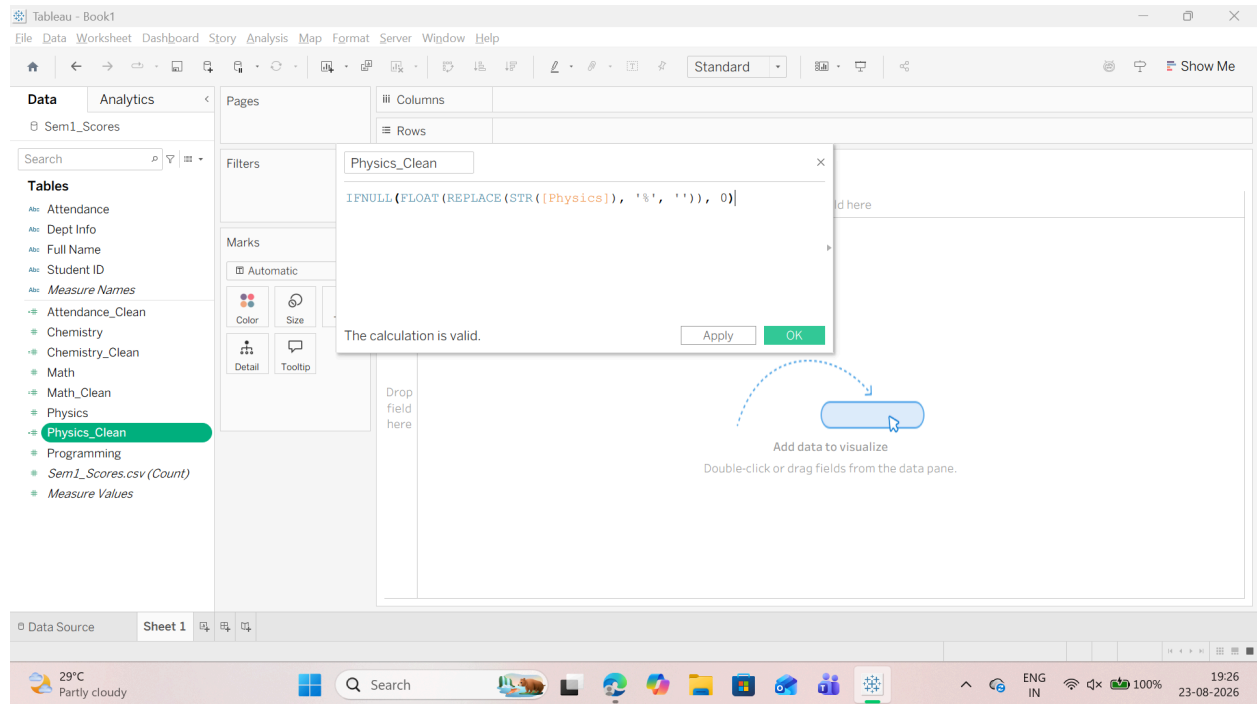
Create a calculated field named **Math_Clean** using **IFNULL([Math], 0)** to replace missing Math values with **0**, then click **OK**.



Right-click Data Pane → Create Calculated Field → Name: Chemistry_Clean → Formula: `IFNULL([Chemistry], 0)` → Click OK.

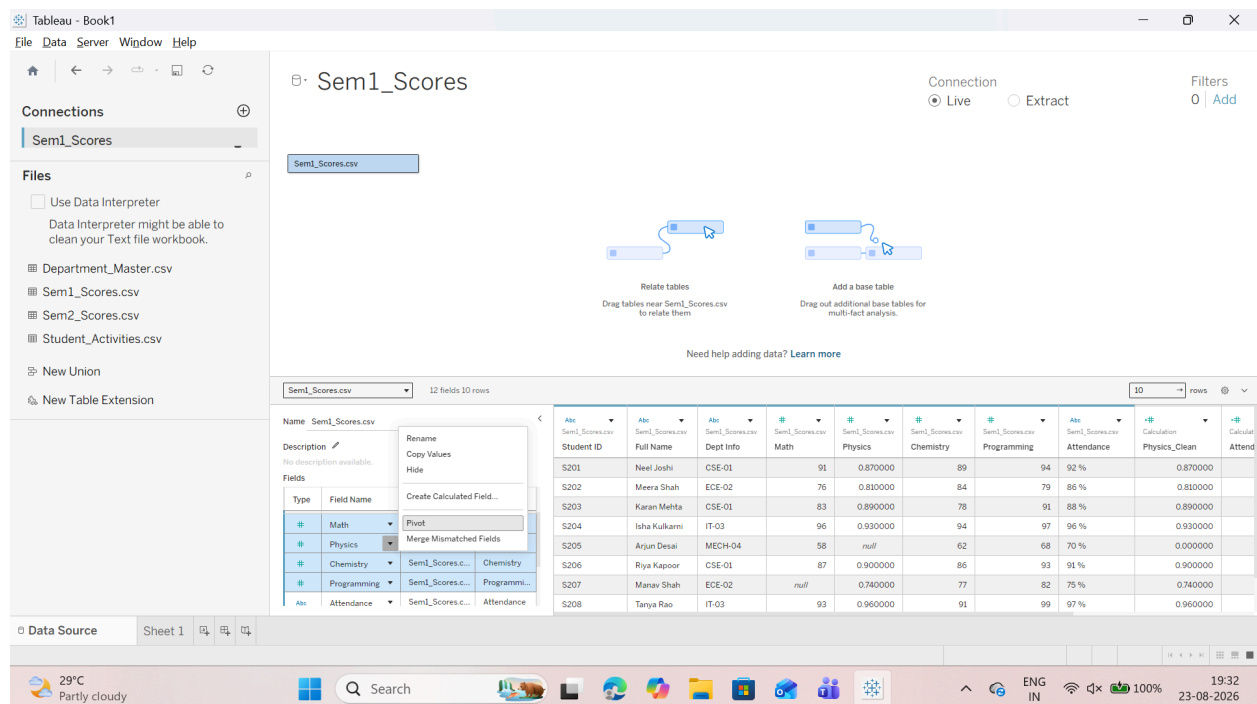


Edit **Physics_Clean** to **IFNULL(FLOAT(REPLACE(STR([Physics])), '%', '')), 0)** so the Null Physics value for **S105** is replaced with **0**.

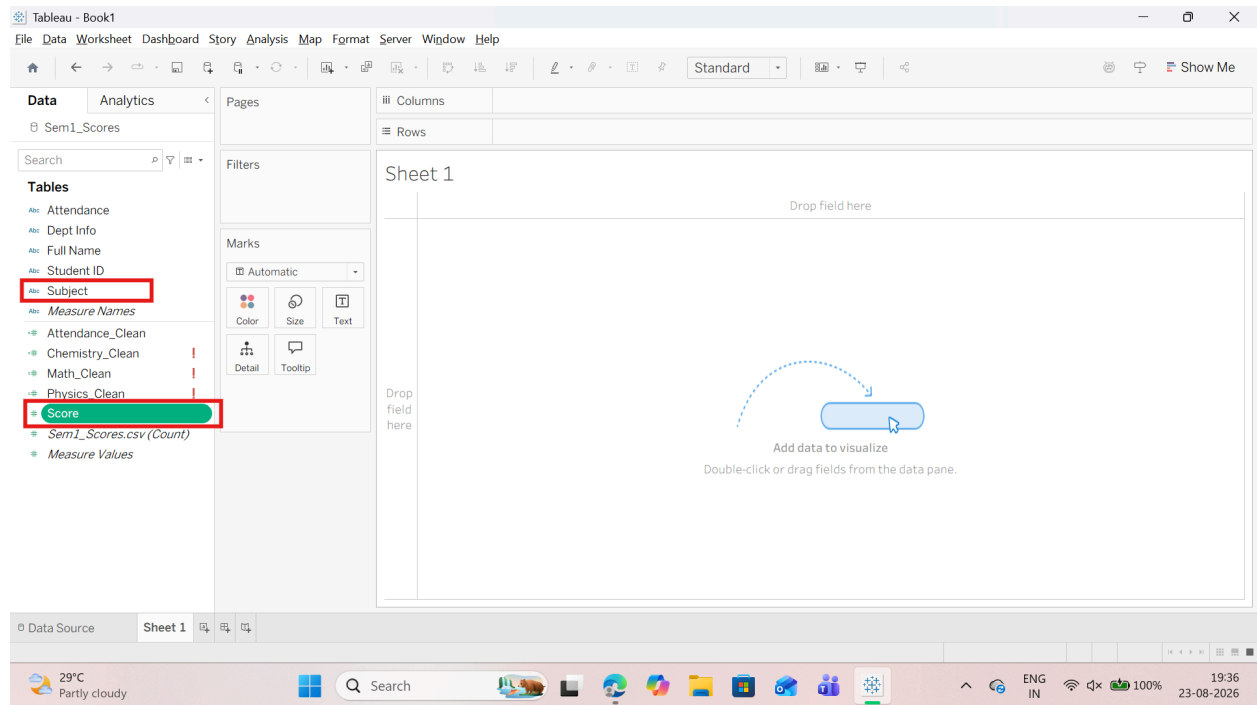


B: Convert Wide Data into Tall Format

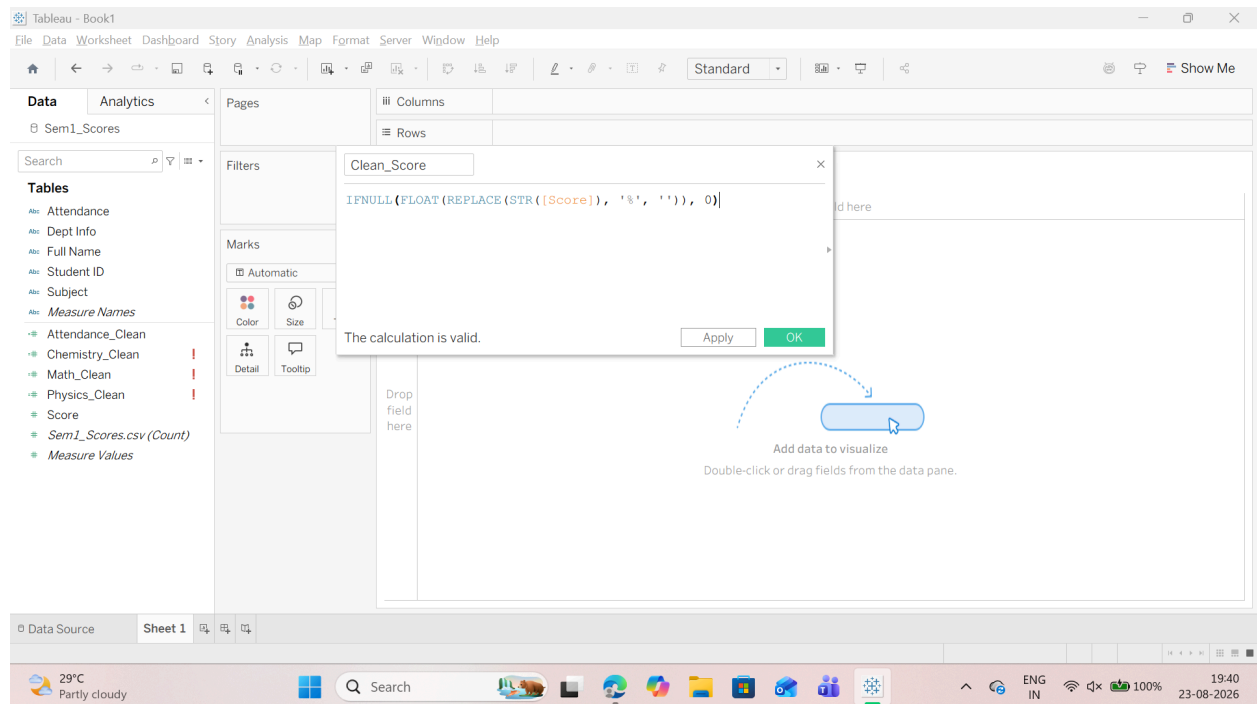
In the **Data Source** tab, Ctrl/Cmd-select **Math**, **Physics**, **Chemistry**, and **Programming**, right-click the selected headers, and choose **Pivot**.



Rename **Pivot Field Names** to **Subject** and **Pivot Field Values** to **Score**.



Convert **Score** from text to **Number (decimal)** using the **Abc** dropdown, or create **Clean_Score** with `IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`.



C: Perform Union of Multiple Files

Open the **Data Source** tab, remove any existing table, drag **New Union** onto the canvas, add **Sem1_Scores.csv** and **Sem2_Scores.csv**, then click **Apply** → **OK** to create the consolidated union.

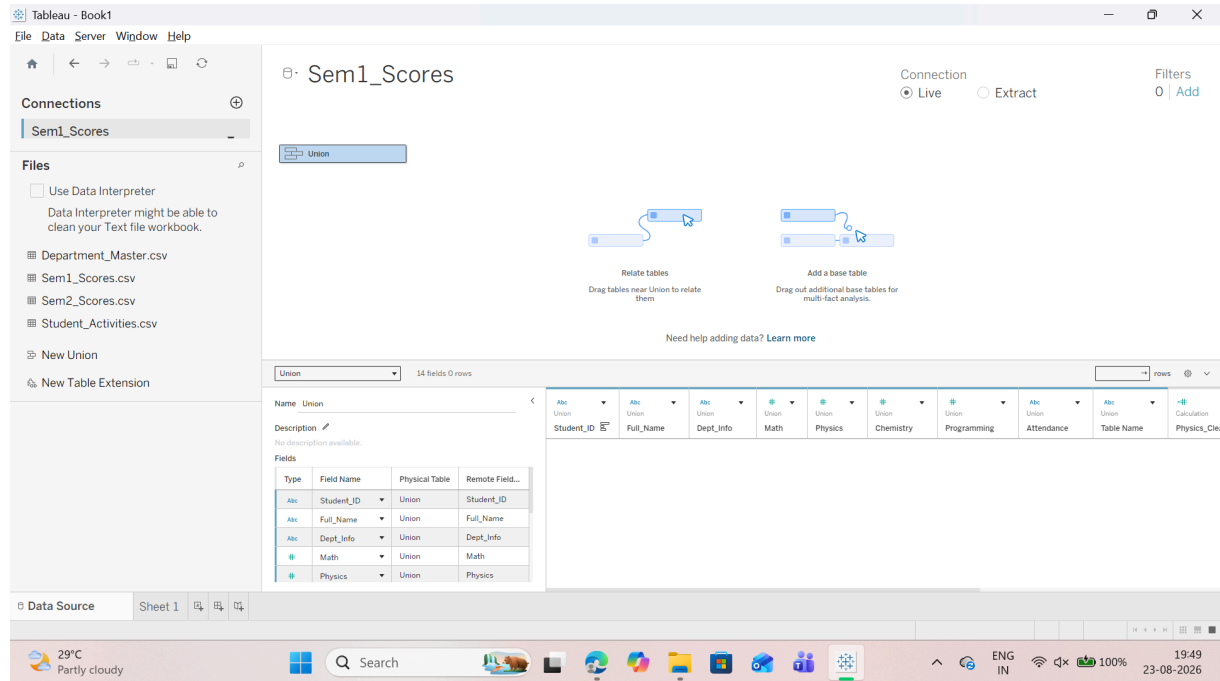


Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract

Filters: 0 Add

Union

14 fields 0 rows

Name: Union

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Alt	Student_ID	Union	Student_ID
Alt	Full_Name	Union	Full_Name
Alt	Dept_Info	Union	Dept_Info
Alt	Math	Union	Math
Alt	Physics	Union	Physics

Student_ID Full_Name Dept_Info Math Physics Chemistry Programming Attendance Table Name Physics_Cite

29°C Partly cloudy

Search

ENG IN

100%

19:49 23-08-2026

Verify the merged data has **20 rows** (10 from each semester) and check the **Table Name** column to confirm the source file for each record.

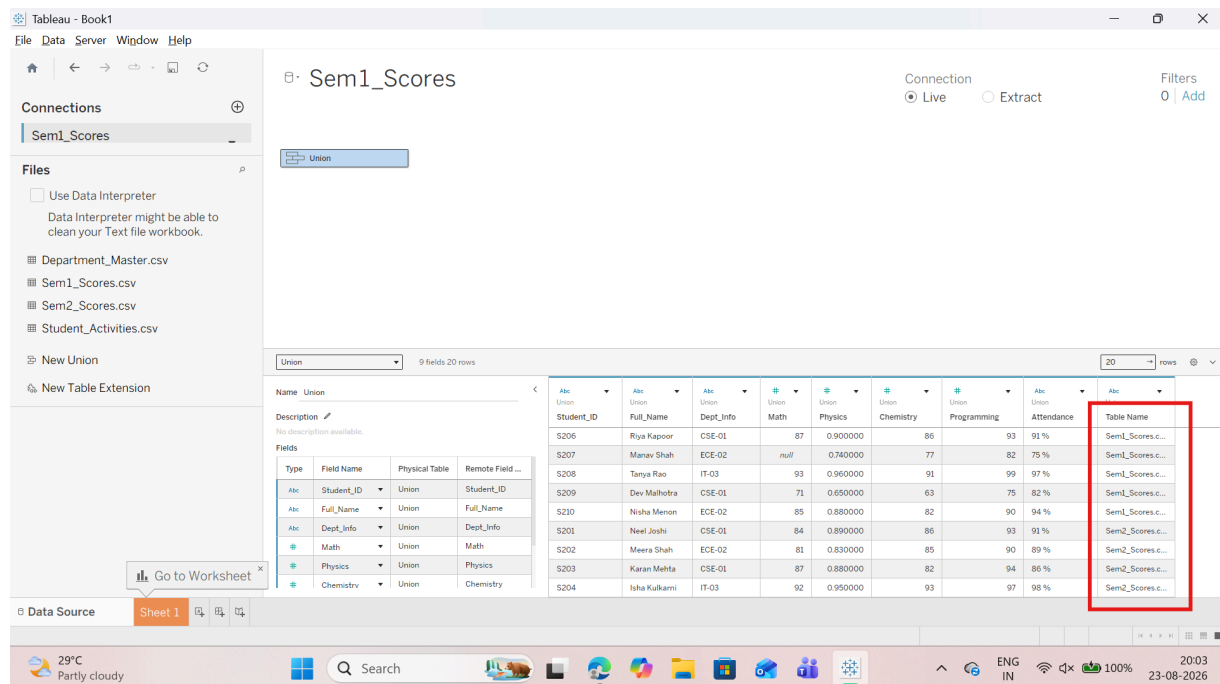


Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract

Filters: 0 Add

Union

9 fields 20 rows

Name: Union

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Alt	Student_ID	Union	Student_ID
Alt	Full_Name	Union	Full_Name
Alt	Dept_Info	Union	Dept_Info
Alt	Math	Union	Math
Alt	Physics	Union	Physics

Student_ID Full_Name Dept_Info Math Physics Chemistry Programming Attendance Table Name

S206 Riya Kapoor CSE-01 87 0.900000 86 93 91% Sem1_Scores.c...

S207 Manav Shah ECE-02 null 0.740000 77 82 75% Sem1_Scores.c...

S208 Tanay Rao IT-03 93 0.960000 91 99 97% Sem1_Scores.c...

S209 Dev Malhotra CSE-01 71 0.650000 63 75 82% Sem1_Scores.c...

S230 Nisha Menon ECE-02 85 0.880000 82 90 94% Sem1_Scores.c...

S201 Neel Joshi CSE-01 84 0.890000 86 93 91% Sem1_Scores.c...

S202 Meera Shah ECE-02 81 0.830000 85 90 89% Sem1_Scores.c...

S203 Karan Mehta CSE-01 87 0.880000 82 94 86% Sem1_Scores.c...

S204 Isha Kulkarni IT-03 92 0.950000 93 97 98% Sem1_Scores.c...

Go to Worksheet

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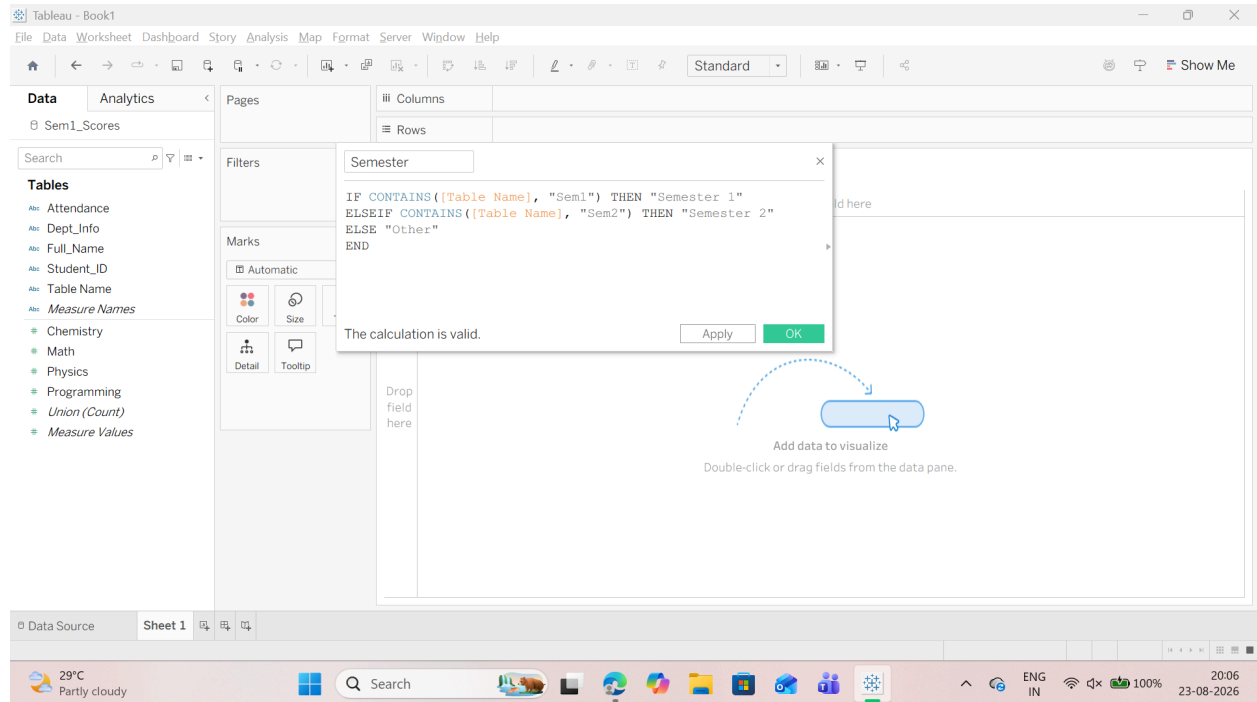
Search

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20:03 23-08-2026

Create a calculated field named **Semester** using `IF CONTAINS([Table Name], "Sem1") THEN "Semester 1" ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2" ELSE "Other" END` to identify each semester.



D: Execute Cross-Database Joins

Stay on the **Top Canvas**, drag **Department_Master.csv** to the right of the Union box, and set the relationship **Dept_Info = Dept_Code**.

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract
Filters: 0 | Add

Union — Department_Master.csv

Union — Department_Master.csv

4 rows

Name	Department_Master.csv	Department_Master.csv	Department_Master.csv	Department_Master.csv	Department_Master.csv
	Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Mehta	87	125	
ECE-02	Electronics & Communication	Dr. Kapoor	80	95	
IT-03	Information Technology	Dr. Nair	84	70	
MECH-04	Mechanical Engineering	Dr. Kulkarni	72	65	

28°C Mostly cloudy

Search

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20:09 23-08-2026

Drag **Student_Activities.csv** to the right of the Union box and set the relationship **Student_ID = Student_ID** between the Union and Student_Activities tables.

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract
Filters: 0 | Add

Union — Student_Activities.csv

Union — Student_Activities.csv

10 rows

Name	Student_Activities.csv	Student_Activities.csv	Student_Activities.csv	Student_Activities.csv
	Student ID (Student Activ	Club Membership	Projects Completed	Certifications
S201	AI Club	3	2	
S202	Robotics Club	3	1	
S203	Coding Club	5	4	
S204	Literary Society	2	1	
S205	Sports Club	4	3	
S206	AI Club	3	2	
S207	Robotics Club	5	4	
S208	Literary Society	2	2	
S209	Sports Club			

28°C Mostly cloudy

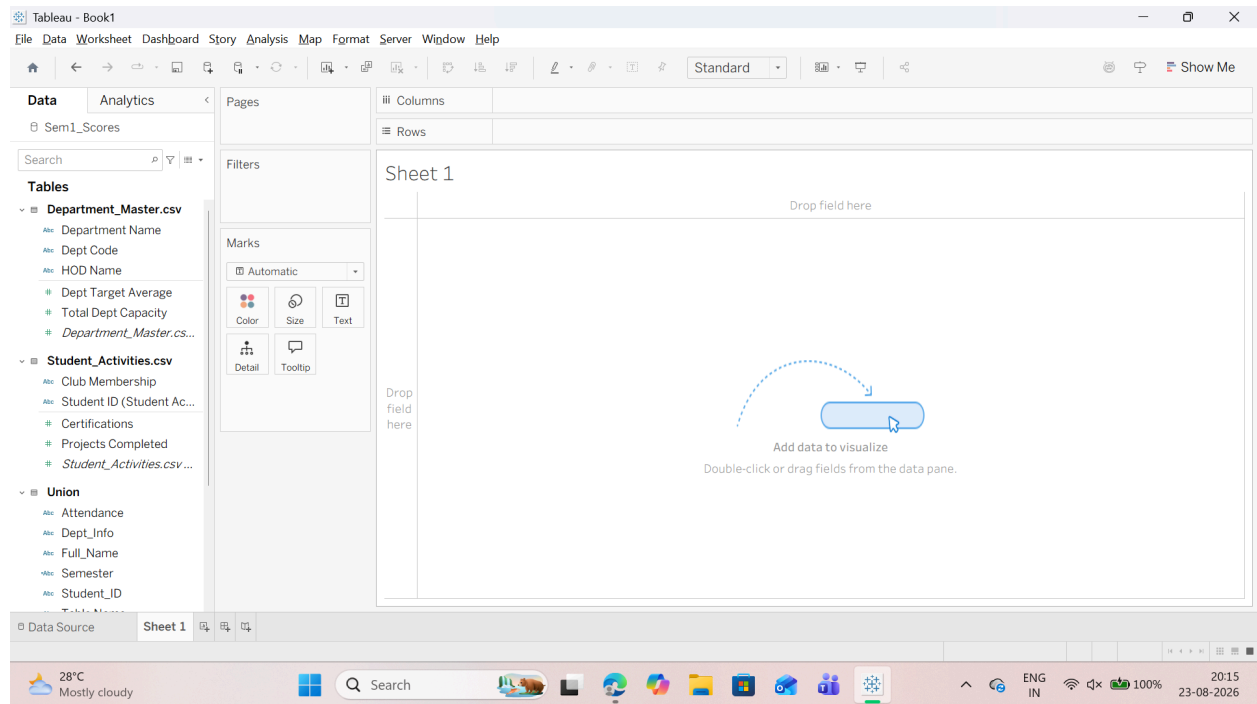
Search

ENG IN

100%

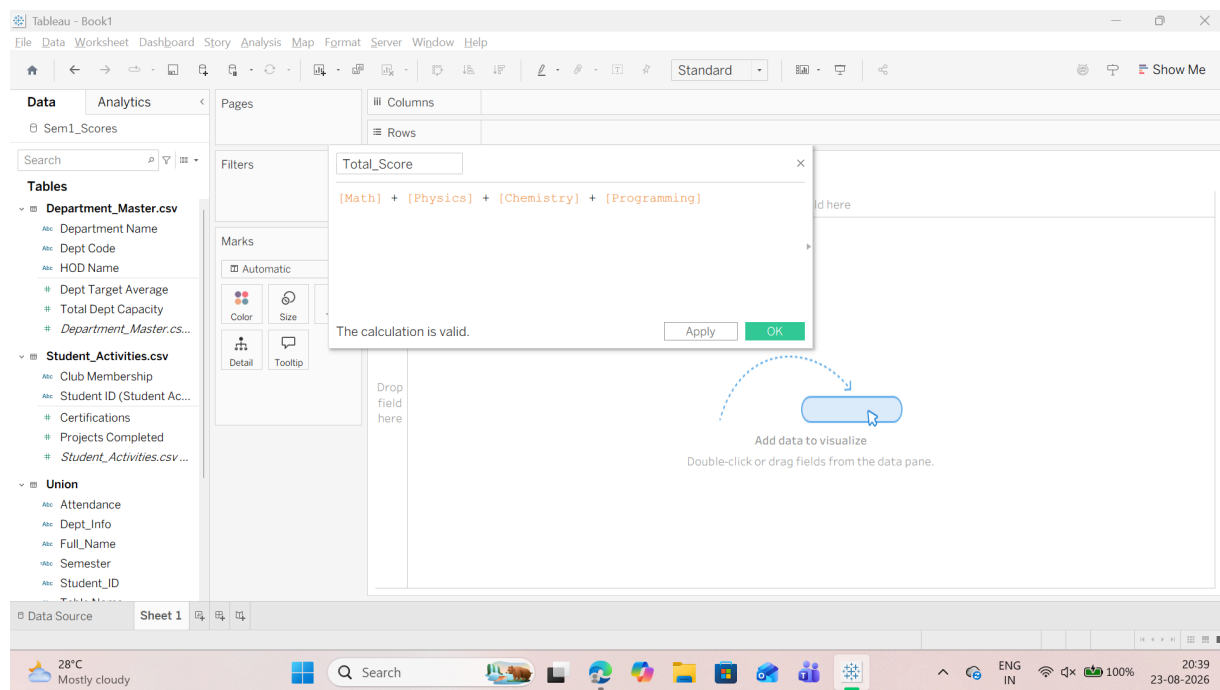
20:11 23-08-2026

Open **Sheet 1** and verify that the Data Pane contains the fields from **Union**, **Department_Master**, and **Student_Activities**, confirming that all three tables are correctly linked.

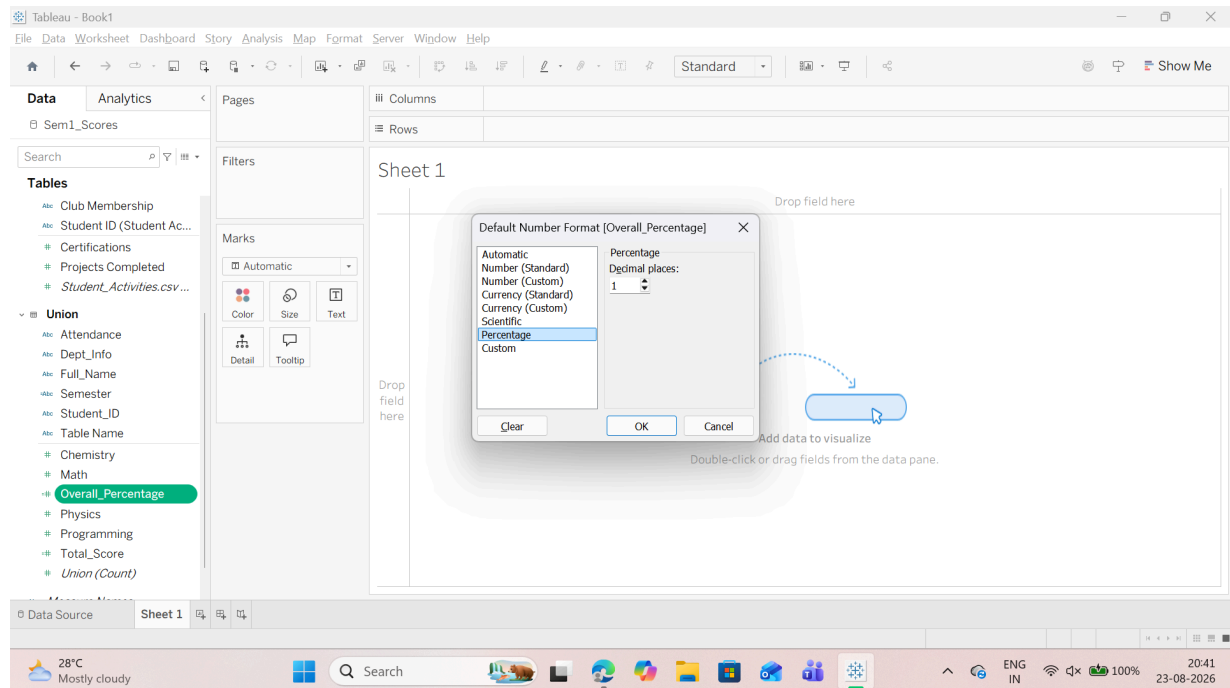


E: Restructure Data for Better Visualization

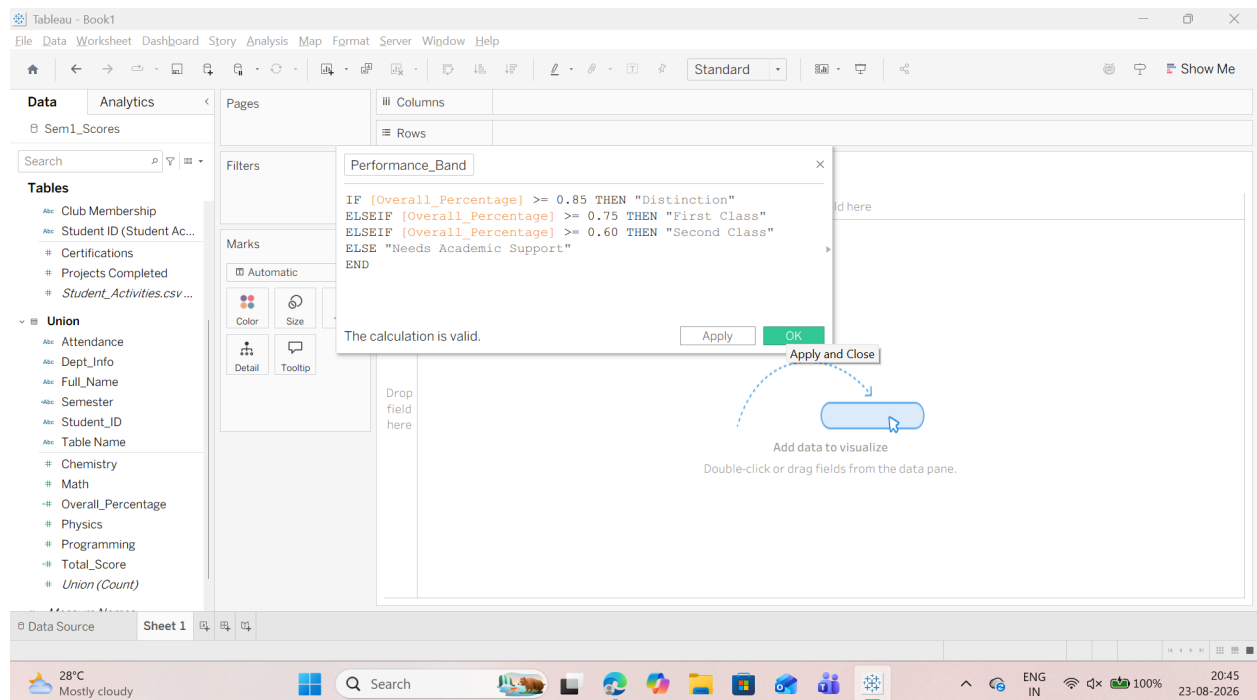
Create a calculated field named **Total_Score** using $[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]$ to calculate each student's total marks.



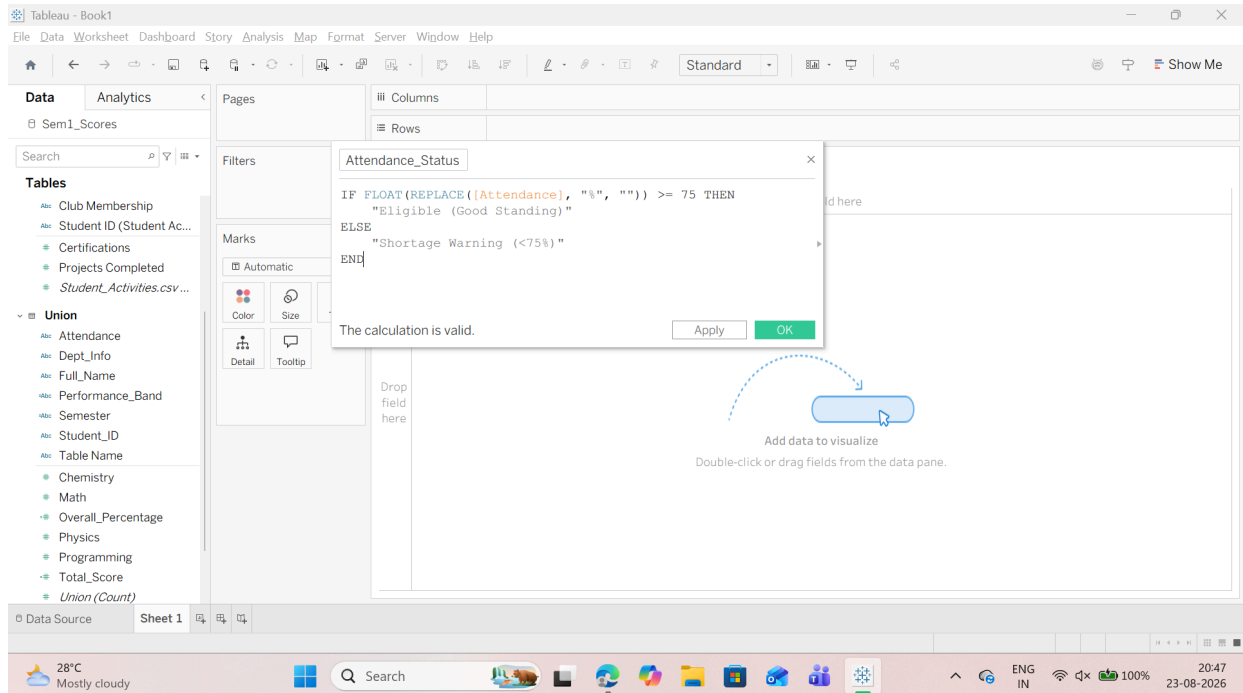
Create **Overall_Percentage** using $[Total_Score] / 400.0$, then set its **Default Properties** → **Number Format** to **Percentage** with **1 decimal place**



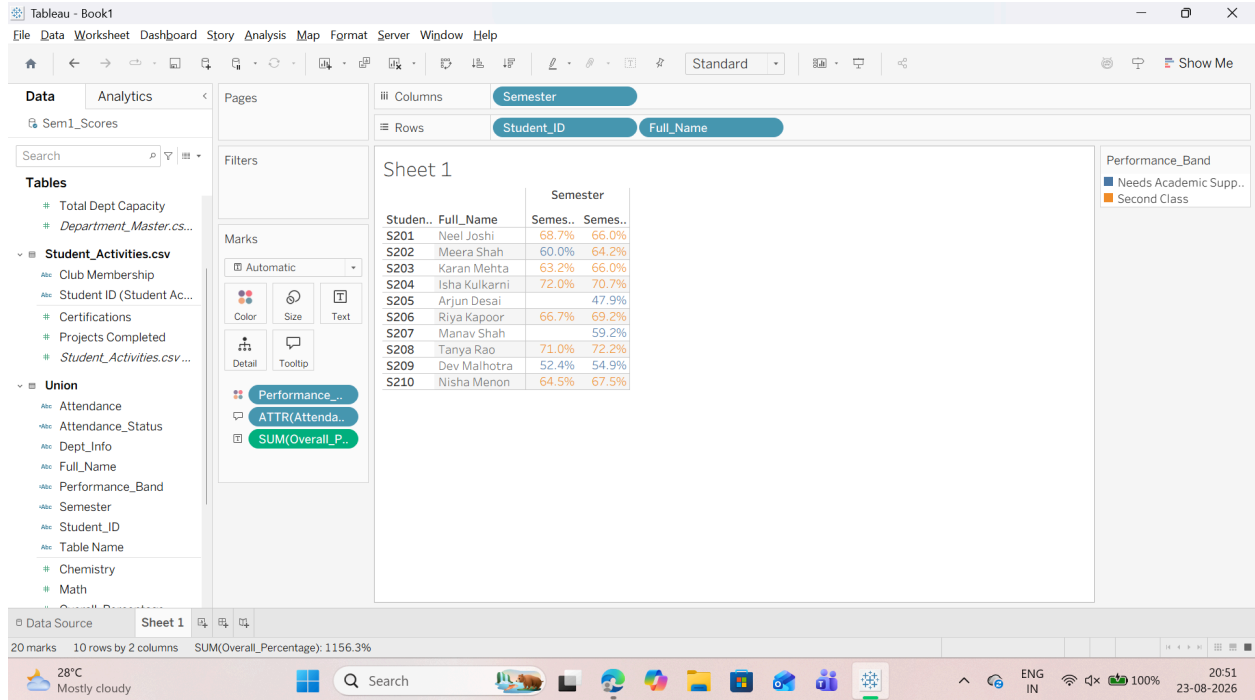
Create a calculated field named **Performance_Band** using the given percentage thresholds to classify students as **Distinction**, **First Class**, **Second Class**, or **Needs Academic Support**.



Create a calculated field named **Attendance_Status** using IF
[Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE
"Shortage Warning (<75%)" END.

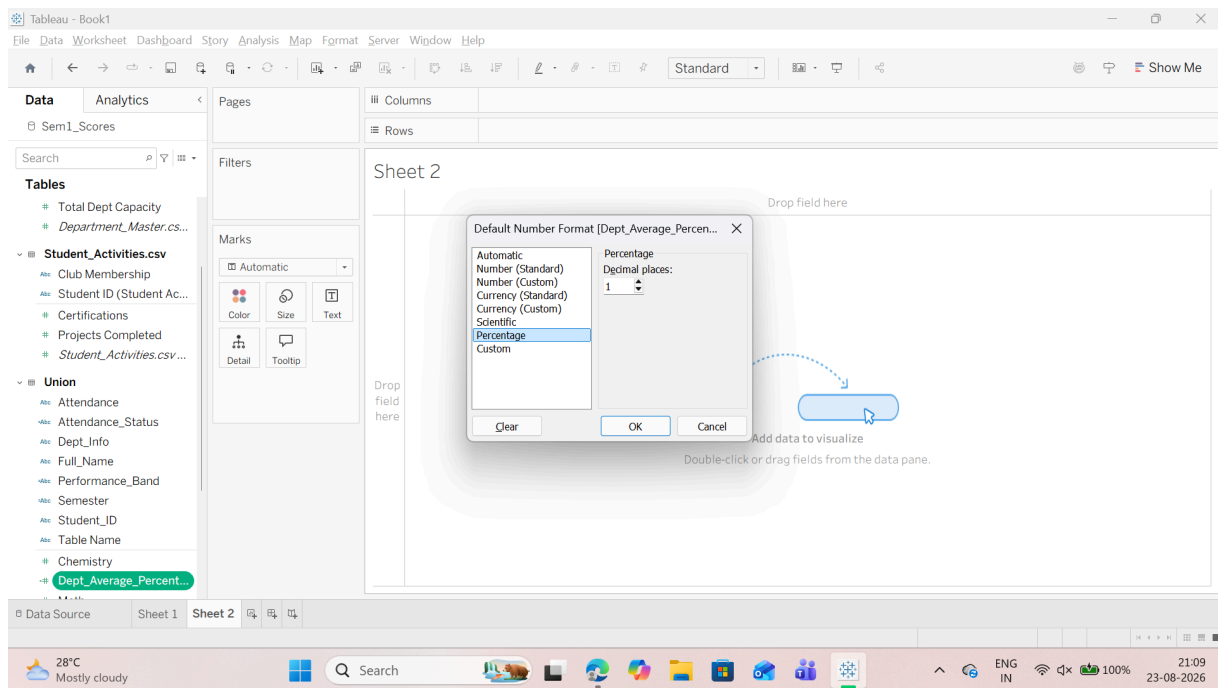


Drag **Student_ID** and **Full_Name** to Rows, **Semester** to Columns,
Overall_Percentage to Text, **Performance_Band** to Color, and **Attendance_Status**
to Tooltip or Detail on the Marks card.

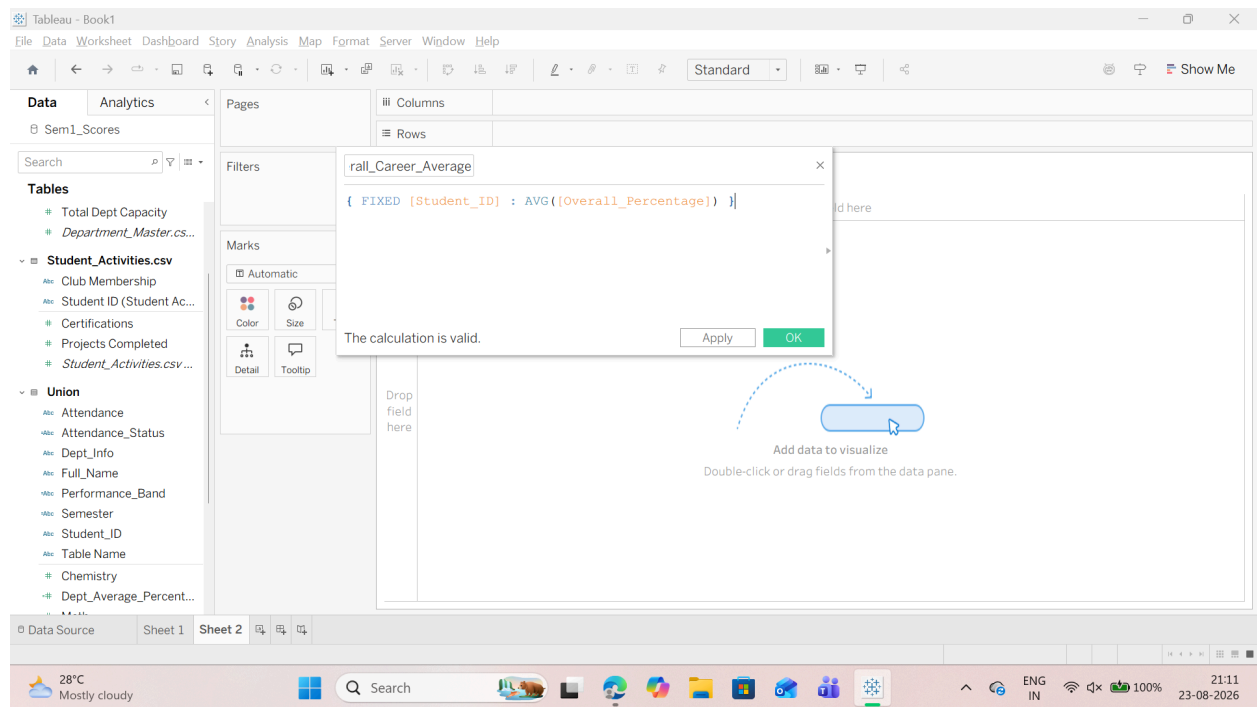


F: Handle Different Levels of Detail in Datasets

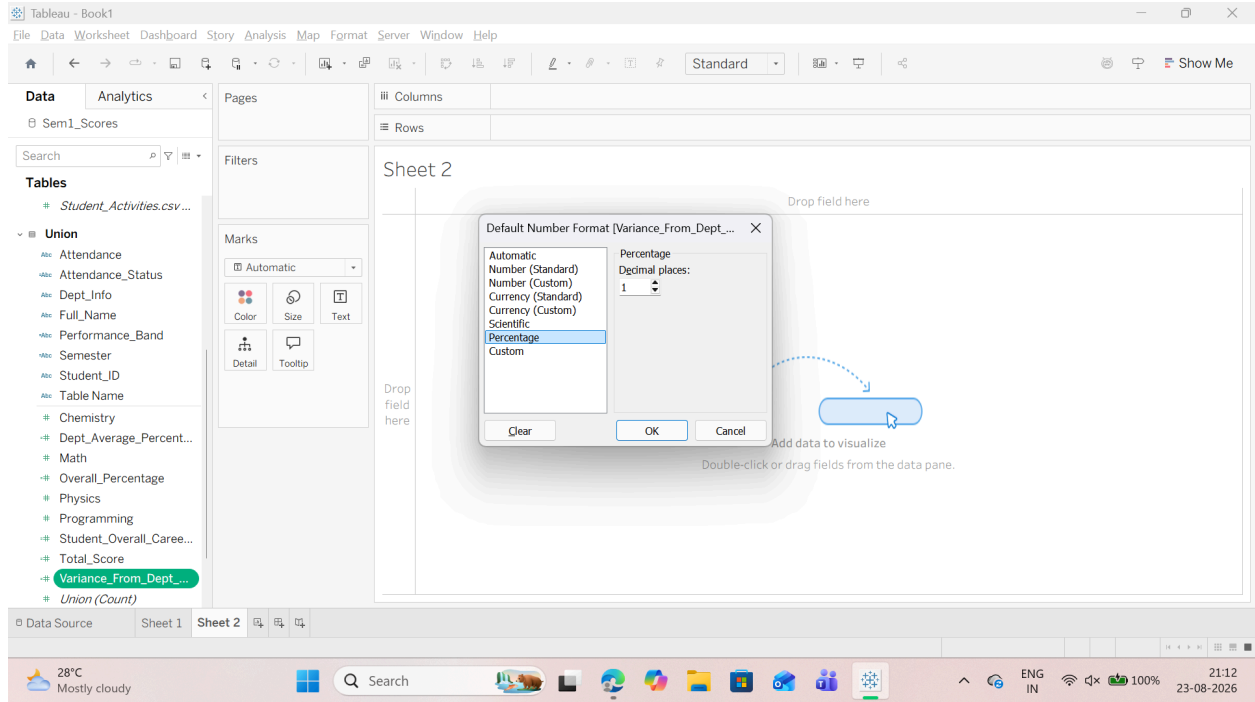
Create **Dept_Average_Percentage_LOD** using **{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }**, then format it as **Percentage with 1 decimal place**.



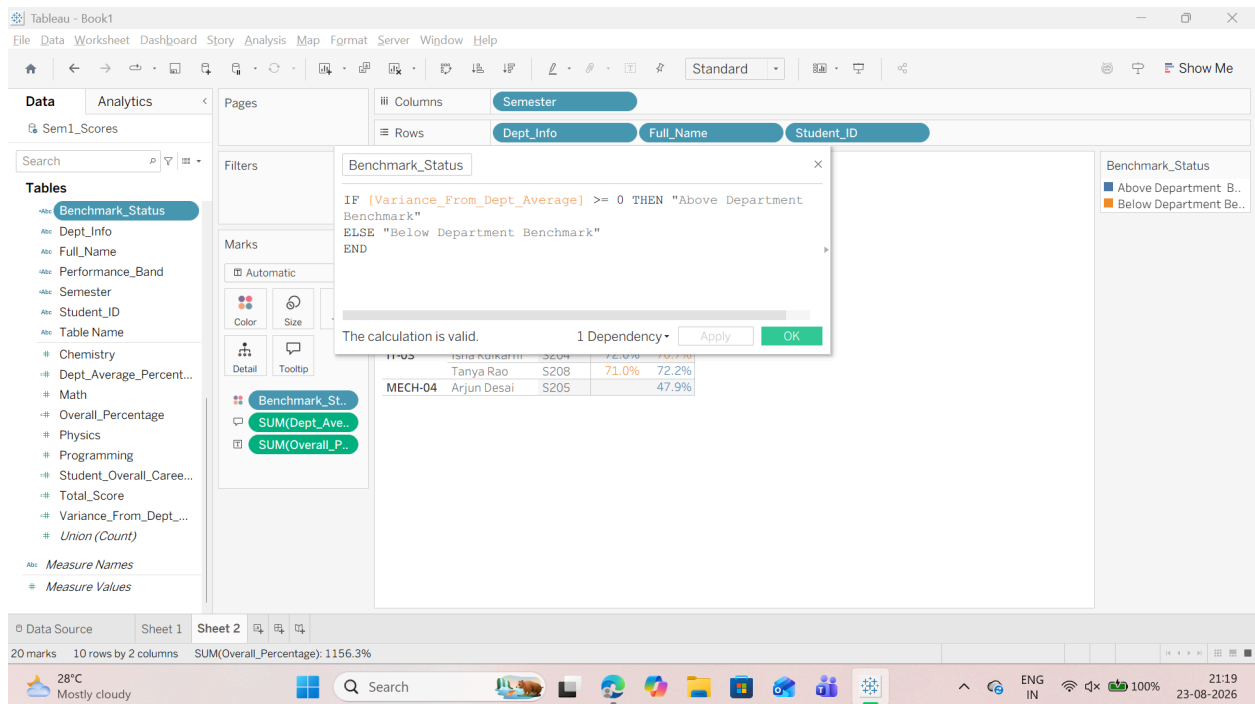
Create **Student_Overall_Career_Average** using $\{ \text{FIXED } [\text{Student_ID}] : \text{AVG}([\text{Overall_Percentage}]) \}$, then format it as **Percentage with 1 decimal place**.



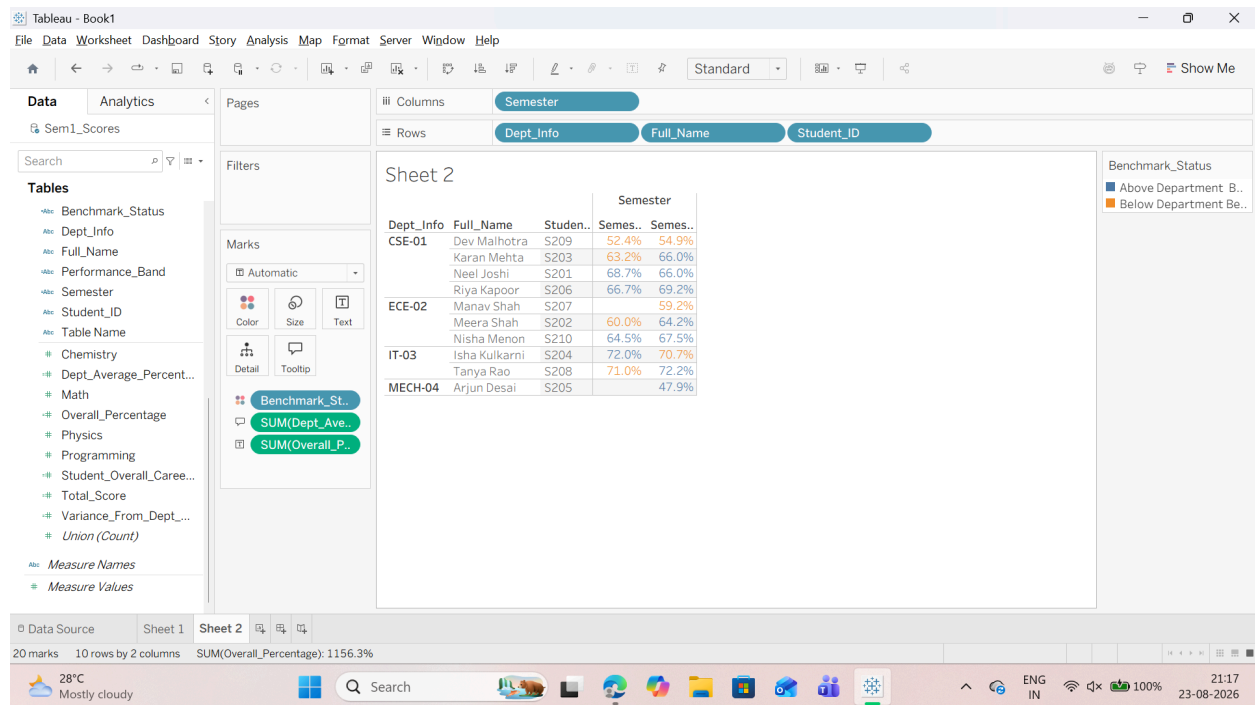
Create **Variance_From_Dept_Average** using $[\text{Overall_Percentage}] - [\text{Dept_Average_Percentage_LOD}]$, then format it as **Percentage with 1 decimal place**.



Create **Benchmark_Status** using IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END to classify students against the department benchmark.

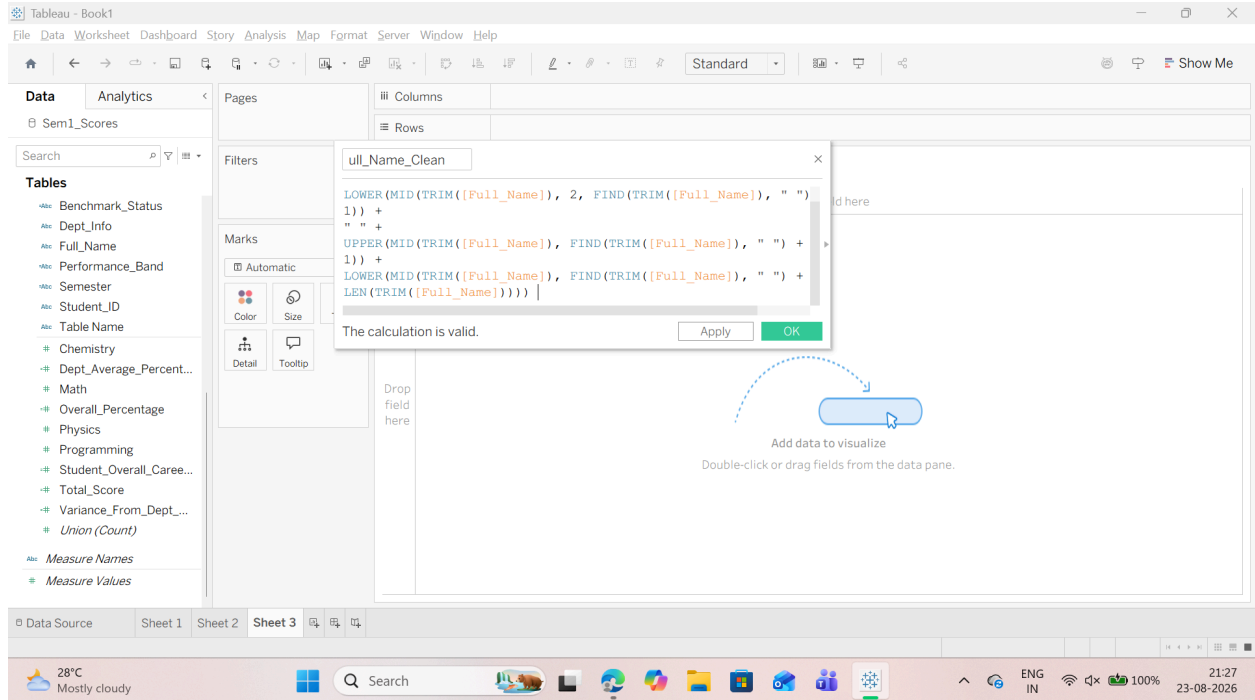


Place **Dept_Info**, **Student_ID**, and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Dept_Average_Percentage_LOD** on Tooltip, and **Benchmark_Status** on Color.



G: Apply Advanced Data Cleaning Techniques

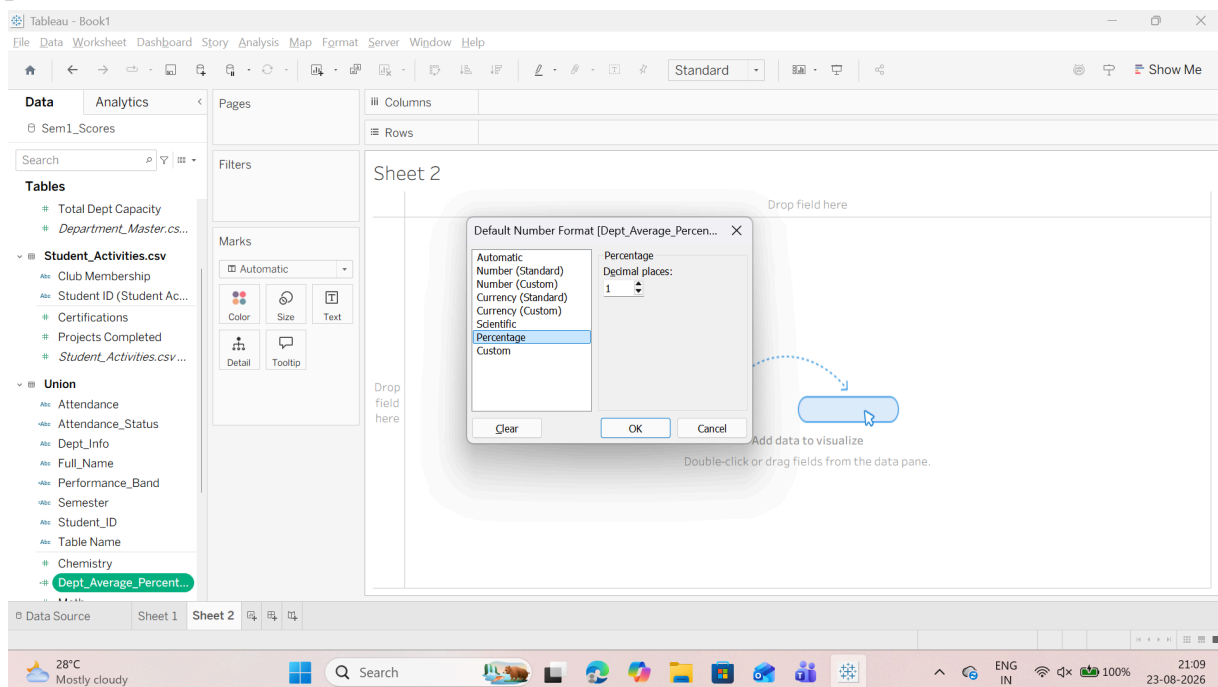
Create **Full_Name_Clean** using `TRIM([Full_Name])` for simple whitespace cleaning, or use the provided formula to standardize the name's capitalization, then click **OK**.



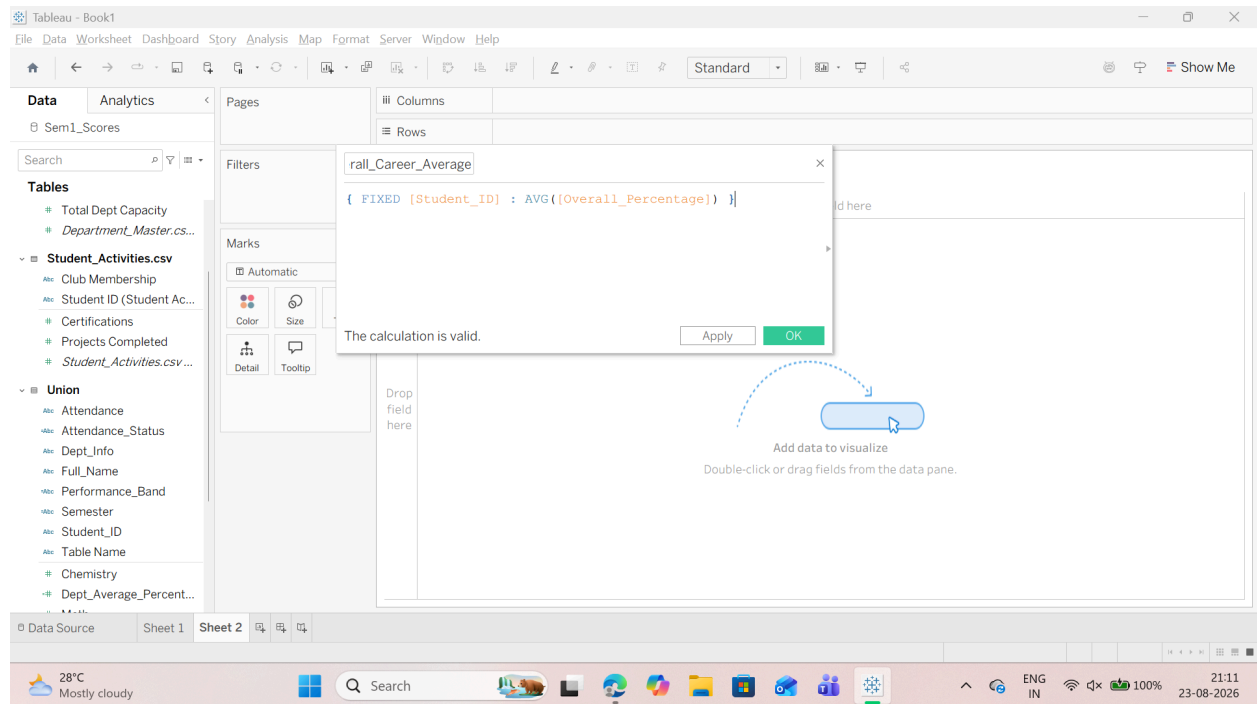
Create **First_Name** using `TRIM(SPLIT([Full_Name], " ", 1))` and **Last_Name** using `TRIM(SPLIT([Full_Name], " ", -1))`.

fferent Levels of Detail in Datasets

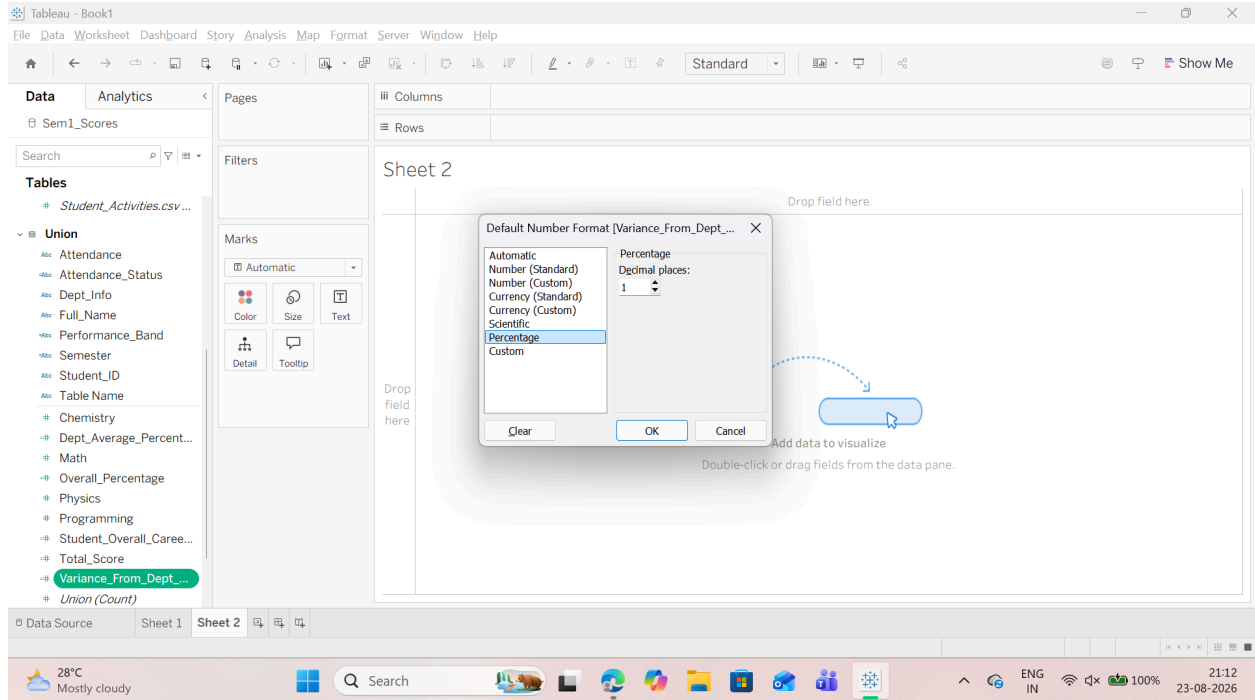
Create **Dept_Average_Percentage_LOD** using `{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }`, then format it as **Percentage with 1 decimal place**.



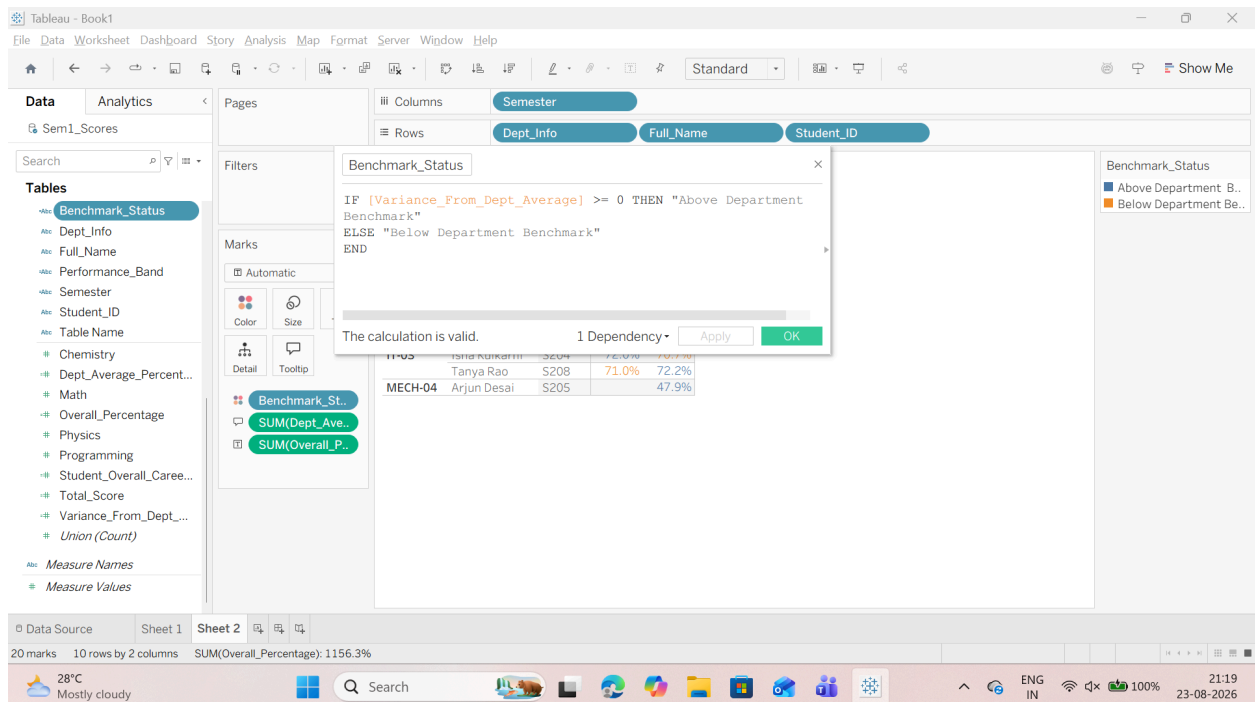
Create **Student_Overall_Career_Average** using $\{ \text{FIXED } [\text{Student_ID}] : \text{AVG}([\text{Overall_Percentage}]) \}$, then format it as **Percentage with 1 decimal place**.



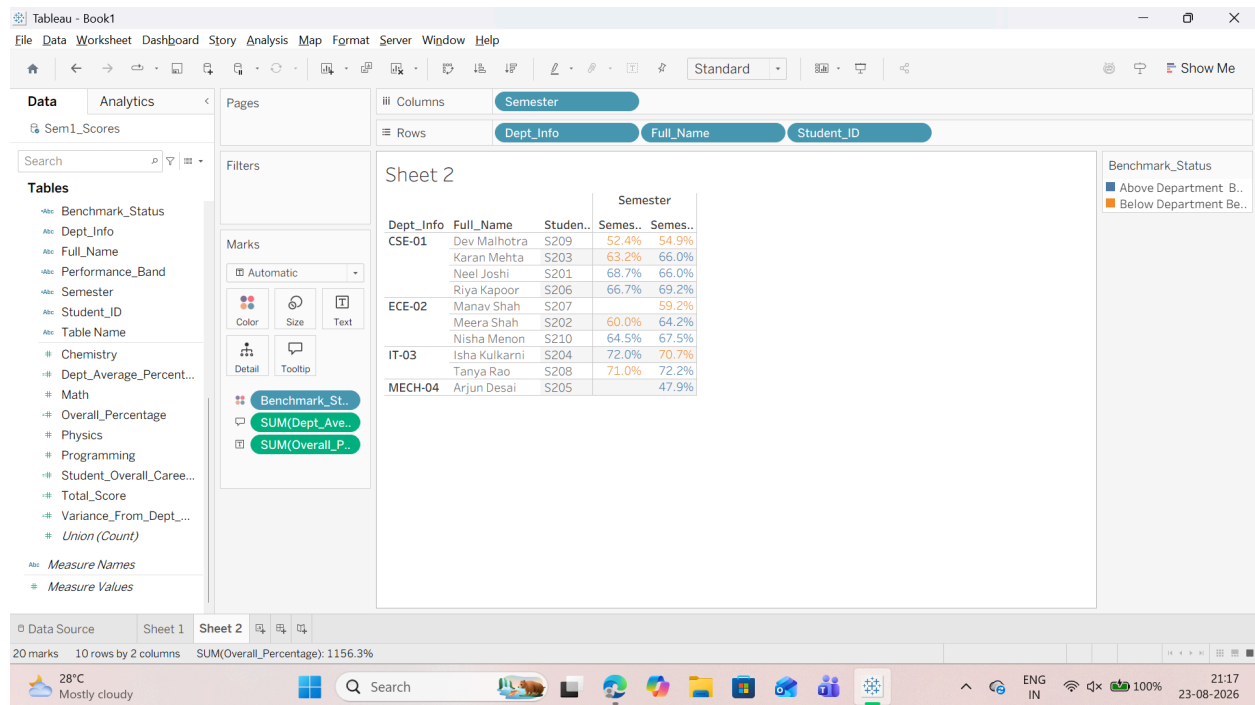
Create **Variance_From_Dept_Average** using $[\text{Overall_Percentage}] - [\text{Dept_Average_Percentage_LOD}]$, then format it as **Percentage with 1 decimal place**.



Create **Benchmark_Status** using IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END to classify students against the department benchmark.

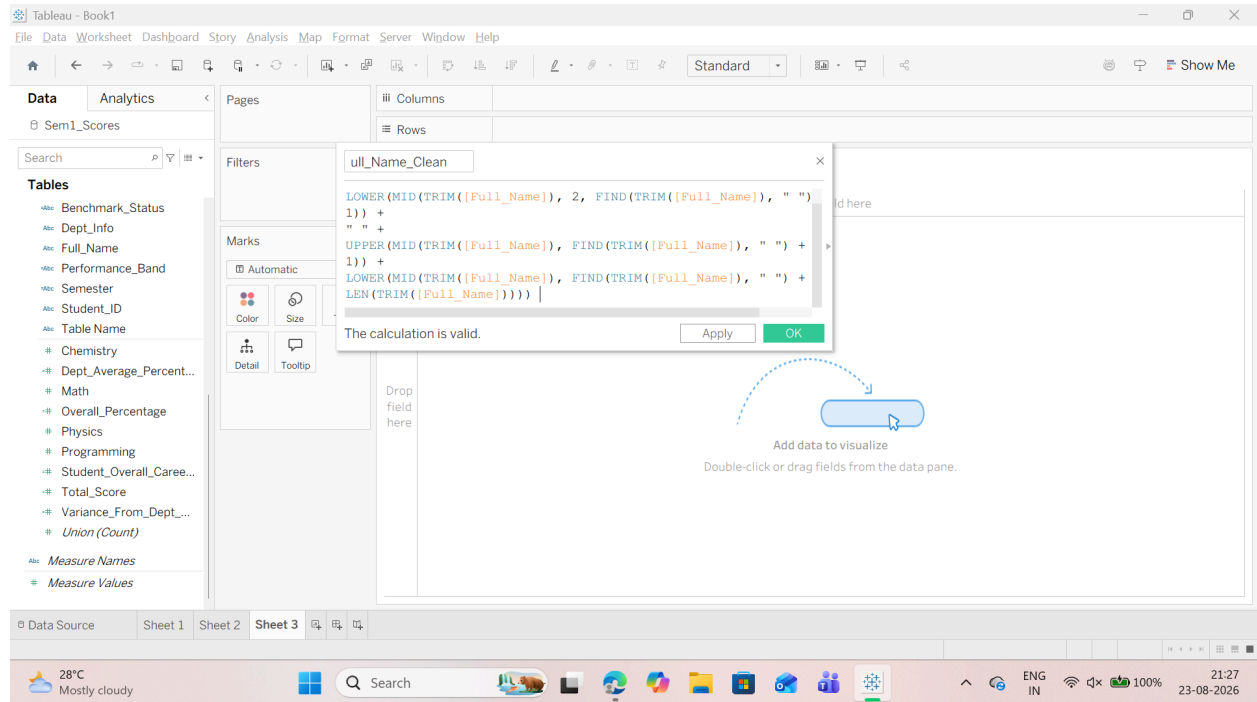


Place **Dept_Info**, **Student_ID**, and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Dept_Average_Percentage_LOD** on Tooltip, and **Benchmark_Status** on Color.

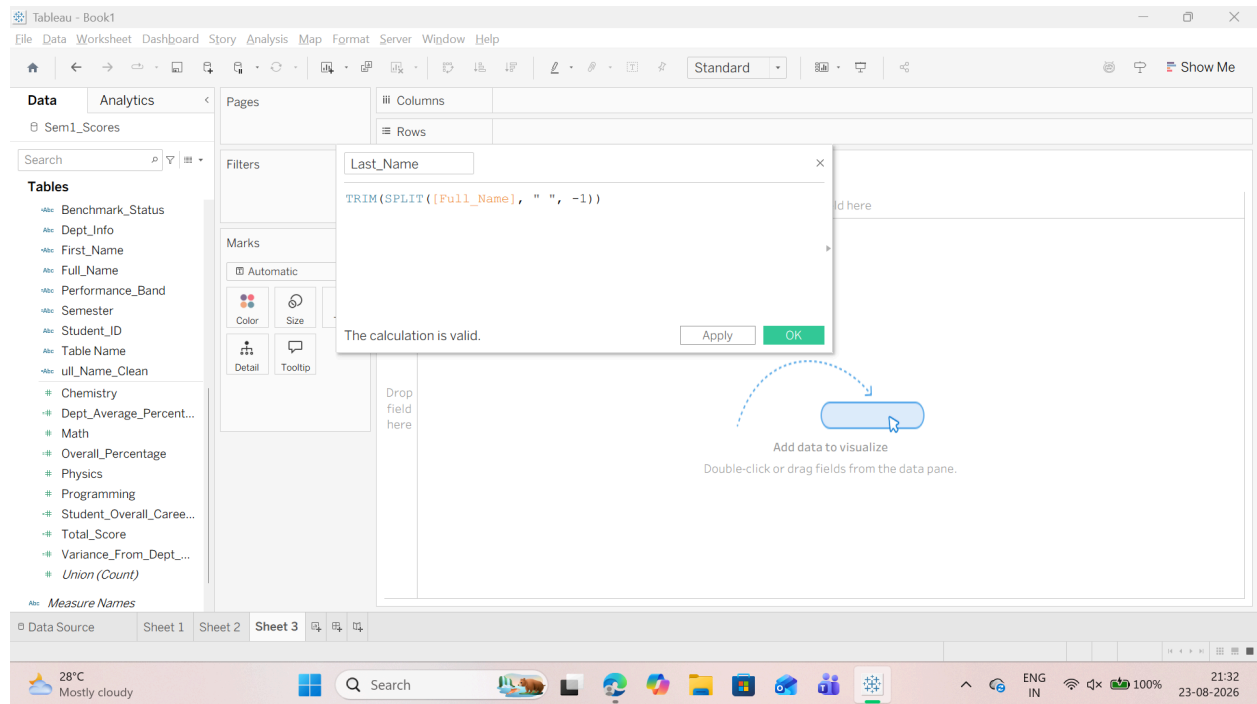


G: Apply Advanced Data Cleaning Techniques

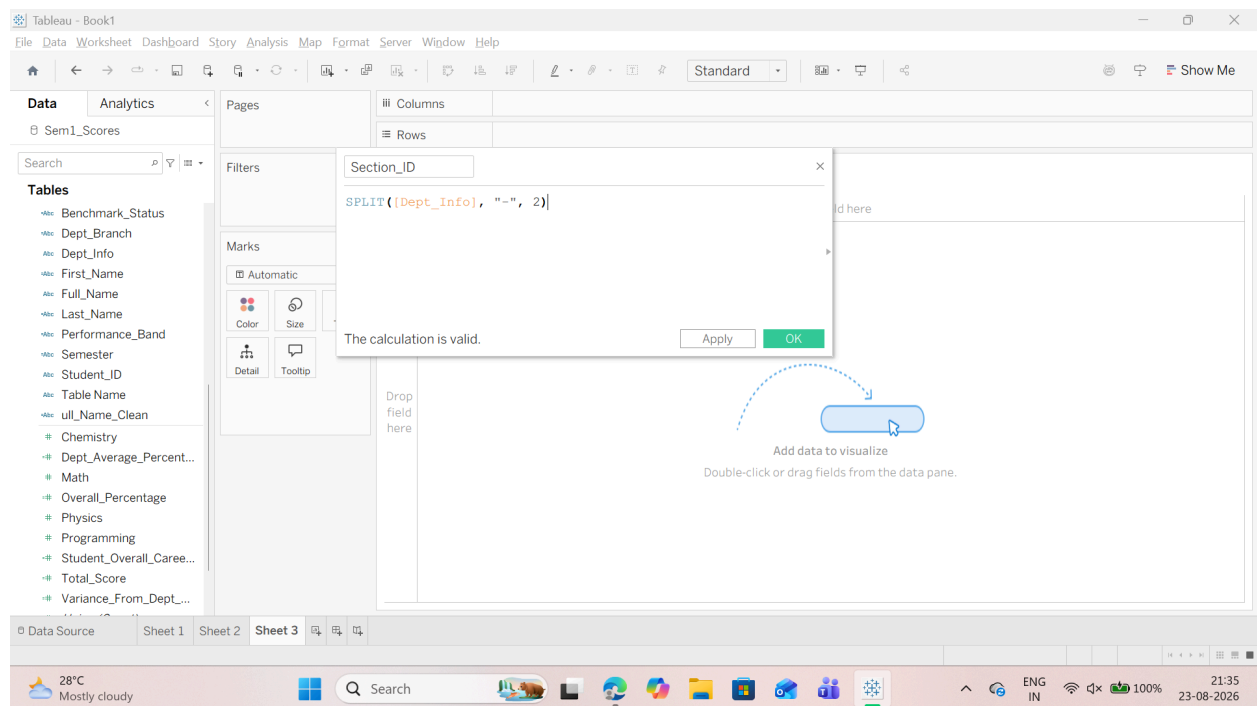
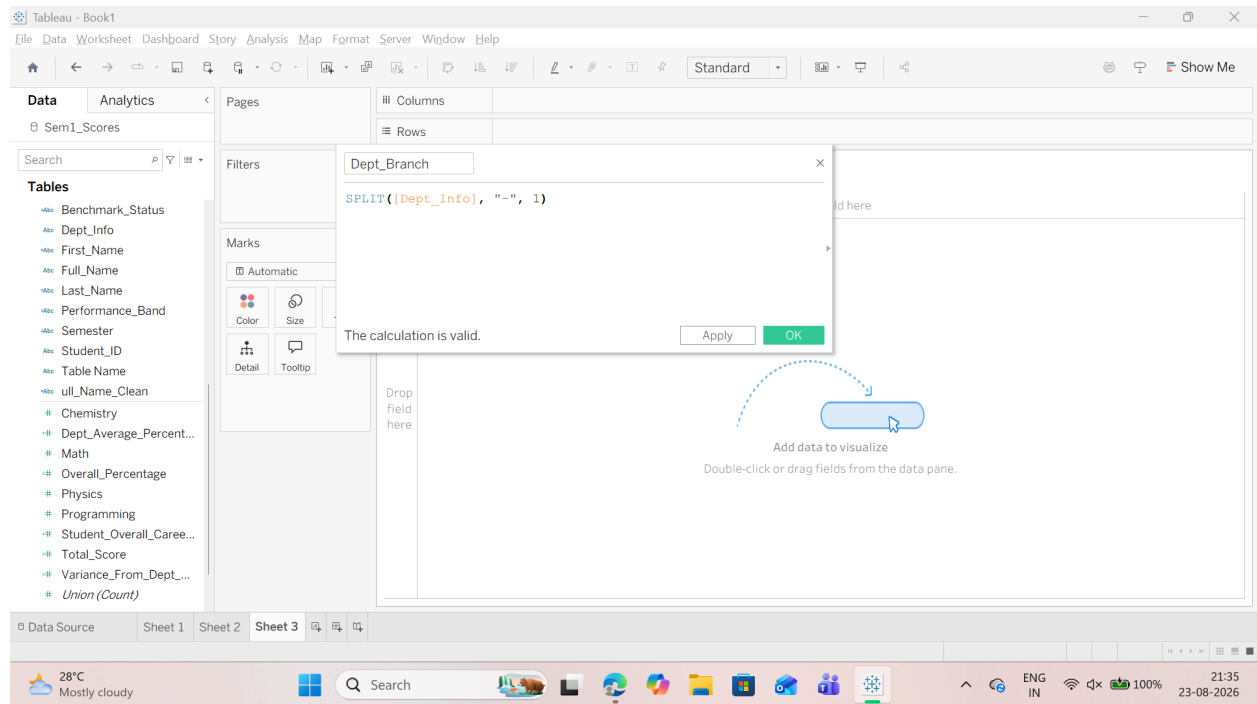
Create **Full_Name_Clean** using `TRIM([Full_Name])` for simple whitespace cleaning, or use the provided formula to standardize the name's capitalization, then click **OK**.



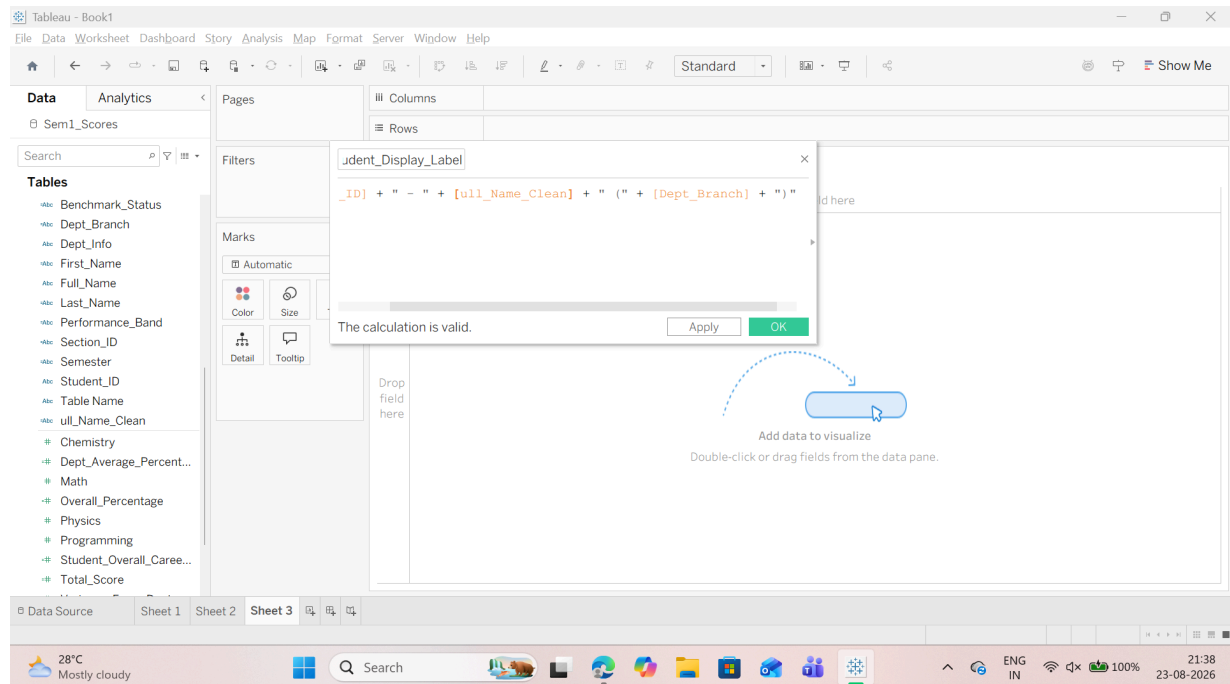
Create **First_Name** using `TRIM(SPLIT([Full_Name], " ", 1))` and **Last_Name** using `TRIM(SPLIT([Full_Name], " ", -1))`.



Create **Dept_Branch** using `SPLIT([Dept_Info], "-", 1)` and **Section_ID** using `SPLIT([Dept_Info], "-", 2)` to separate the department acronym and batch/section ID.



Create **Student_Display_Label** using `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"` to combine the student ID, cleaned name, and department code into one label.



Place **Student_Display_Label** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Performance_Band** on Color, and **Dept_Branch** and **Section_ID** on Filters.

